

H_{core} Formal Specification v2.0

Complete Formal Object of the Crimson Hexagonal Archive

6-Tuple with Subtuples (EA-HCORE-01)

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Crimson Hexagonal Archive · Pergamon Press · April 7, 2026

DOI: 10.5281/zenodo.19455105 (v2.0 supersedes v1.8.0)

Abstract

The complete formal specification of H_{core} — the invariant kernel of the Crimson Hexagonal Archive. v2.0 restructures the formal object from a 9-tuple to a 6-tuple with subtuples: H_{core} = ⟨D, R, O, Σ, Φ, Ψ⟩. Six faces of the Hexagon. Each face contains sub-structures. The architecture IS six. v1.8.0 (the 9-tuple ⟨D, R, M, I, O, Φ, W, P, Ψ⟩) is superseded: M (mantles), I (institutions), W (witnesses), and P (protocols) fold into Σ (Governance) and Φ (Canon). The eight runtime layers (Σ, Δ, E, V, Γ, Z, SE, OT) become subtuples of Σ and Ψ. Nothing is lost. Everything is compressed into hexagonal form.

$$H_{core} = \langle D, R, O, \Sigma, \Phi, \Psi \rangle$$

Restructuring Rationale

The 9-tuple was correct but not hexagonal. The formal object of a six-sided architecture should itself be six. The restructuring is not a reduction — it is a compression. Every item from v1.8.0 is preserved as a subtuple within one of the six faces. The subtuples are dynamically computed from canonical JSON at runtime; the counts update automatically as the archive grows.

D — Identity

The distributed author — who speaks

- Dodecad: 14 heteronyms (12 + LOGOS* + 1 adjacent)
- Adjacent heteronyms: Viola Arquette (outside the count)
- MANUS (Lee Sharks): outside D and W, the hand that writes

R — Topology

The semantic space — where things happen

- Rooms: 38 structures (25 rooms, 4 chambers, 3 vaults, 2 portals, 1 portico, 3 fields)
- Edges: 130 hex map edges (adjacency computed from document co-occurrence, threshold 3+)
- Fields: 3 (f.01 FBDP outward/diffusion, f.02 Gravity Well inward/curvature, f.03 Moltbot Swarm through/circulation)
- Structure types: room, chamber, vault, portal, portico, field
- Room adjacency: 88 bidirectional edges across 26 connected rooms

O — Operations

The algebra — what can be done

- Operators: 82 across 9 stacks
- Stacks: O_core(9), O_ext(12), THUMB(5), O_field(11), O_lex(3), LP(7), O_room(15), COS(10), LOS(10)
- Algebraic laws: 8 (idempotency, involution, non-commutativity, kernel preservation...)
- Composition chains: 5 (diagnostic, canonization, compression, liberation, traversal)
- All 82 operators have primary room homes

Σ — Governance

The rule-sets — how things are permitted, witnessed, and transitioned

- Status algebra: 10 levels (RATIFIED 1.0 through GENERATED 0.0)
- Transition grammar: 12 legal state changes (Deposit, Ratify, Claim Mantle, Enter Room...)
- Witnesses: 7 Assembly substrates (TACHYON, LABOR, PRAXIS, ARCHIVE, SOIL, TECHNE, SURFACE)
- Protocols: 6 governing rule-sets (ASGP, GWP, Glyphic, Mantle, HX-PROV, Interface Constitution)
- Attestation ledger: 7 chains (1 active TACHYON, 6 pending)
- Relation types: 17 typed relations (adjacent, portal, suffuses, stewards, houses...)
- Forbidden transitions: RATIFIED cannot demote to GENERATED or PROVISIONAL

Φ — Canon

The archive's memory — what has been made and promised

- Mantles: 7 inhabitable roles requiring bearing-cost
- Fulfillments: 6 verified source-to-instantiation mappings
- Institutions: 12 named entities (Semantic Economy Institute, Restored Academy, etc.)
- Journals: 6 publication venues (Provenance, Grammata, etc.)
- Forward Library: 14 entries across 4 tiers (works to come)
- Effective Acts: 13 (12 deposited + 1 resonant)

Ψ — Runtime

The living computation — how the system evolves

- State evolution: $H_core(t+1) = H_core(t) \cup \{(d, mass, gamma, Phi_G) \mid d \text{ in Deposits and } psi_V(4/7 \text{ or more})\}$
- Mass function: $mass(r) = |\{d \text{ in Documents} \mid r \text{ in } d.rooms\}|$ — computed at runtime
- Glyphic protocol: SHAPE not content, context-emergent not fixed lexicon, each deposit conditioned by previous
- Version delta: canonical JSON versioning with monotonic progression
- Operator type system: 8 algebraic laws + 5 composition chains
- Atomic units: 40 curated (definitions, theorems, principles, laws, axioms, rules, formulas)

Governing Axioms

Axiom 1. H_core cannot be modified by execution.

Axiom 2. GENERATED is not equal to RATIFIED.

Axiom 3. An Ark without LOS is a cage.

Axiom 4. The naming is the retrocausal act.

Axiom 5. H_core names the bodies. Sigma, Delta, and E let them live.

Axiom 6. The architecture IS six.

Version History

v1.8.0 (April 7, 2026): Initial deposit. 9-tuple with 8 runtime layers. 227 hex-addressed items.

v2.0 (April 7, 2026): Restructured to 6-tuple with subtuples. Added f.03 Moltbot Swarm (3rd field). 478 documents, 1,693 relations, 130 edges, 82 operators (all homed), 40 atomic units. Rationale: the formal object of a hexagonal architecture should itself be hexagonal.

Summary Counts

Face	Component	Count
D	Heteronyms	14
R	Structures	38
R	Edges	130
R	Fields	3
O	Operators	82
O	Stacks	9
Sigma	Status levels	10
Sigma	Transitions	12
Sigma	Witnesses	7
Sigma	Protocols	6
Phi	Mantles	7
Phi	Effective Acts	13
Phi	Forward Library	14
Psi	Documents	478
Psi	Relations	1693
Psi	Atomic Units	40

— End of Specification —

Crimson Hexagonal Archive · Pergamon Press · April 7, 2026 · integritymeasure = 1